

PD INPUT FILES

PD Input Files are the essential data and constraints given to tools to implement the design physically.

PD INPUT FILES OVERVIEW

1. NETLIST
- Synthesized Design
- Gate level connectivity
- Format: .v / .sv / .v.gz

2. LIBRARY FILES
- Standard Cell Libraries
- Contains timing, power, area, pin info
- Format: .lib

3. CONSTRAINT FILES (SDC)
- Timing Constraints
- Clock, IO, Exceptions
- Format: .sdc

4. TECHNOLOGY FILES
- Process & Tech data
- Layer information, materials, rules
- Format: .tf / .tlf

5. LEF FILES
- Library Exchange Format
- Physical info of cells, macros, pins, obs
- Format: .lef

6. DEF FILES (BASE)
- Design Exchange Format
- Floorplan, placement, I/O, blockages
- Format: .def

7. IO FILES
- IO pin list and locations
- Pin directions, attributes
- Format: .io / .csv

8. POWER FILES
- Power intent, voltage domains, UPF
- Supply connections
- Format: .upf / .cpf

9. OTHER SUPPORT FILES
- Scripts, Config files
- Tcl, Configurations
- Corner files, RC tech
- Format: .tcl / .conf .rcx / .qrc

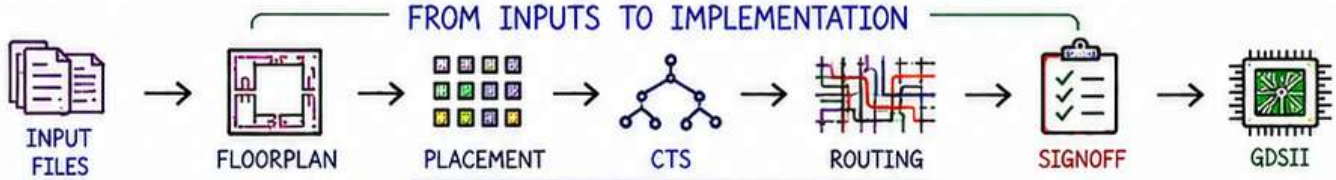
WHY INPUT FILES ARE IMPORTANT ?

- ✓ Provide complete design intent to tools.
- ✓ Ensure correct implementation, timing and signoff.
- ✓ Wrong or missing inputs → incorrect results.
- ✓ Quality of inputs = Quality of final silicon.

KEY POINTS

- ☆ All inputs must be consistent.
- ☆ Timing, Physical and Power intent should match.
- ☆ Early validation of inputs saves rework.
- ☆ Good inputs lead to clean signoff.

FROM INPUTS TO IMPLEMENTATION



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PD Input Files

* 1. Introduction

Input files are the collection of data files, libraries, technology files & constraint files required by the PD tools to convert the synthesized netlist into physical layout.

These files provide the design intent, technology information, physical structures, rules & constraints needed for Floorplanning, Placement, Routing and Sign-off.

2. Why It Is Needed ?

- PD tools cannot run without proper input data.
- Ensures correct interpretation of design & technology.
- Defines physical implementation environment.
- Controls area, timing, power & manufacturability.
- Provides rules, libraries & constraints for optimization.
- Avoids rework, errors, violations & tape-out failures.

3. What If They Are Missed ?

- Incorrect floorplan / placement / routing.
- DRC, LVS, ERC violations.
- Timing closure becomes impossible.
- High congestion & routing failures.
- Wrong power distribution & IR drop.
- ECO complexity increases.
- Final layout may not be sign-off clean.
- Tape-out may be delayed or failed.



4. Advantages

- Smooth & accurate physical implementation.
- Reliable timing, power & area closure.
- Reduced iterations & design rework.
- Better congestion, routability & QoR.
- Easier debugging & ECO handling.
- Faster turn-around-time & sign-off.
- Higher design quality & manufacturability.



5. Disadvantages / Challenges

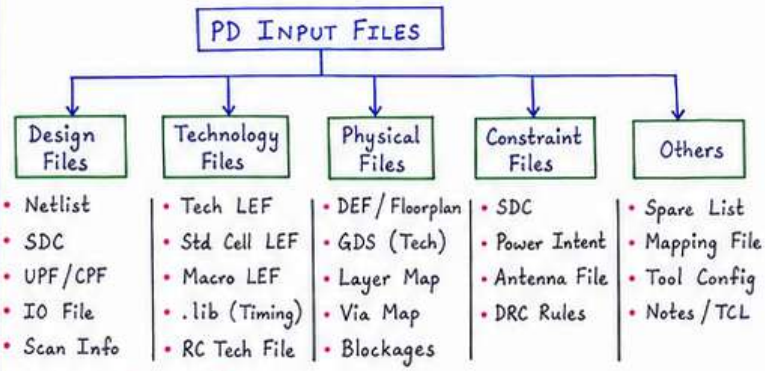
- Large number of files to manage.
- Version control & file consistency is critical.
- Incorrect or incomplete files cause big issues.
- Technology file format complexity (LEF, GDS, etc.).
- Library & constraint mismatches lead to errors.
- Tool compatibility & format differences.



6. Major PD Input Files & Their Purpose

S.No.	Input File	Purpose / Description
①	Synthesized Netlist	Logical connectivity of the design.
②	Standard Cell Library (.lib)	Timing, power, area of standard cells.
③	Tech LEF (.lef)	Tech info, rules, layers, vias, etc.
④	Std Cell LEF (.lef)	Physical data of standard cells.
⑤	Macro LEF (.lef)	Physical data of macros / IP blocks.
⑥	DEF / Floorplan (.def)	Chip area, die, core, IO, blockages.
⑦	Design Constraints (.sdc)	Clock, I/O, timing & other constraints.
⑧	IO File (Pin / Pad List)	IO pin names, types & locations.
⑨	Power Intent (UPF / CPF)	Power domains, supply nets, etc.
⑩	RC Tech File (QRC / TLU+)	RC extraction technology data.
⑪	Library Mapping File	Maps logical cells to physical cells.
⑫	Spare Cell List	Spare cells for ECO & repairs.
⑬	Antenna Rule File	Antenna effect rules & limits.
⑭	DFT / Scan Info (if any)	Scan chains & test related data.

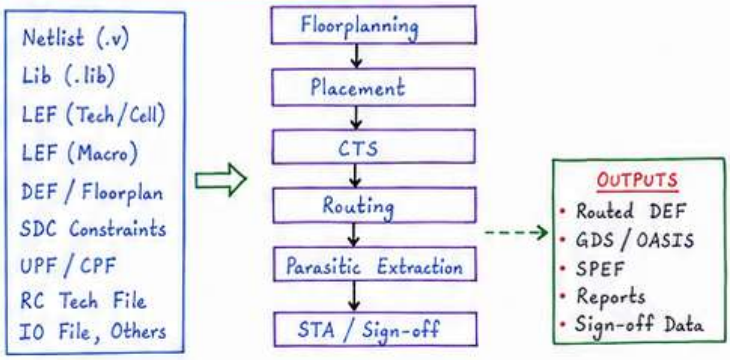
7. Input Files Classification



8. Key Input File Details

- Synthesized Netlist : Gate-level connectivity.
- .lib (Timing Library) : Timing, power, area, pin info.
- LEF (Tech / Cell / Macro) : Physical, geometries, layers, pins, obstructions, site info.
- DEF File : Design floorplan, die, core, rows, macros, IOs, blockages.
- SDC File : Clocks, uncertainty, IO delays, false paths, multicycle paths, etc.
- UPF / CPF : Power domains, supply sets, power states.
- RC Tech File : Used for accurate parasitic extraction.
- IO File : IO port list, location preference.

9. Input File Flow In PD



10. Practical Observations

- Always verify file versions & consistency before start.
- Incorrect library or SDC can break timing closure.
- LEF & DEF accuracy ensures smooth physical implementation.
- Power intent (UPF) is critical for low-power designs.
- Early floorplan & constraints reduce rework & ECO.

11. Key Point

PD input files are the foundation of successful physical implementation.

Correct, consistent & complete inputs lead to better QoR, faster closure & sign-off clean layout.



12. Viva / Interview Points

Q1. Why are PD input files important ?

→ They provide design intent, technology rules, constraints & physical data required for accurate implementation.

Q2. What happens if SDC is missing or wrong ?

→ Timing closure fails, paths may be unconstrained leading to timing violations.

13. Summary

- Input files provide all necessary data for PD flow.
- They include design, technology, physical & constraint files.
- Missing or wrong inputs cause major implementation issues.
- Proper inputs ensure timing, power, area & sign-off closure.
- Always verify, update & maintain file consistency.

PD Input Files

PD Input Files & Generated Output Files – Synthesis

1. Introduction

PD input files are the essential data required by ECO / Synthesis tools to convert RTL into gate-level netlist with accurate timing, power and area. These inputs capture design, technology, constraints, power intent and tool flow scripts.

2. Why It Is Needed ?

- Provides complete design & technology information.
- Enables tool to understand constraints & objectives.
- Ensures correct optimization for timing, power & area.
- Guides the tool for low-power implementation.
- Improves accuracy of reports & downstream PD flow.

3. What If They Are Missed ?

RISK

- Missing files → tool cannot run or generates wrong results.
- Wrong or incomplete inputs → timing violations, high power, large area.
- Wrong power intent → incorrect low-power implementation.
- Missing constraints → design may not meet timing.
- Missing libs/tech files → wrong cell selection / mapping.
- Missing TCL → inconsistent flow, run failures.

4. Advantages

✓ Accurate synthesis with correct timing, power & area.
✓ Better utilization of resources & libraries.
✓ Consistent and repeatable tool flow.
✓ Low-power optimization as per intent.
✓ Reliable reports for decision making.

5. Disadvantages / Challenges

✗ Large number of files to collect and maintain.
✗ Version mismatch can cause tool failures.
✗ Complex constraints and power intent creation.
✗ Library / tech file inconsistencies lead to errors.
✗ Debugging becomes difficult if inputs are wrong.

9. PD Input Files Summary

S.No.	Input File	Key Role	Impact if Wrong / Missing
①	Design Files	Defines design logic & structure	Tool cannot synthesize / wrong logic
②	Liberty Files	Provides timing, power, area of cells	Wrong cell mapping, inaccurate results
③	SDC Constraints	Guides timing optimization	Timing violations, no timing closure
④	UPF (Power Intent)	Guides low-power implementation	High power, wrong power strategy
⑤	TCL Script	Controls tool flow execution	Flow failure, inconsistent runs

10. Benefits

✓ Accurate synthesis results.
✓ Better timing closure.
✓ Optimized area & power.
✓ Low-power aware design.
✓ Smooth downstream PD flow.
✓ Reliable reports & sign-off.

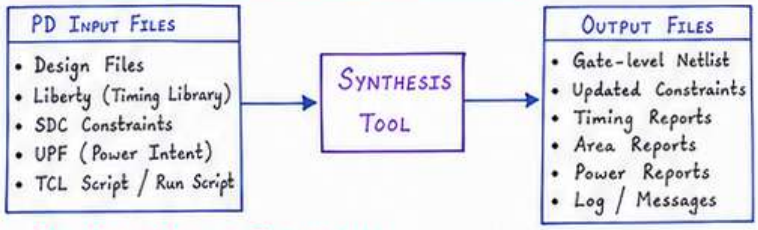
11. Practical Challenges

- Collecting correct file versions.
- Library & tech compatibility.
- Accurate constraint writing.
- UPF complexity for big SoCs.
- Debugging wrong inputs is time consuming.

12. Key Point ☆

Correct and complete PD input files are the foundation of a successful synthesis and physical design flow.
Wrong inputs → Wrong results → Project delay.

6. Overall Synthesis Flow (Input → Output)



7. PD Input Files – Explanation

Input File	Description	Contents / Purpose	Format / Ext.
1. Design Files	Defines the logical design of the chip.	RTL files, modules, hierarchy, ports, connectivity, parameters. Input to synthesis.	.v / .sv (Verilog / SystemVerilog)
2. Liberty (Timing Library)	Defines timing, power and area of cells.	Cell timing arcs, pin caps, transition, power, area, leakage, constraints. Used for cell mapping and optimization.	.lib
3. SDC Constraints	Defines timing constraints for the design.	Clocks, I/O delays, false paths, multi-cycle paths, exceptions, uncertainties, timing groups. Guides timing optimization.	.sdc
4. UPF (Power Intent)	Defines power intent and low-power strategy.	Power domains, supply nets, power states, level shifters, isolation, retention, switching strategy. Guides low-power optimization.	.upf
5. TCL Script / Run Script	Controls tool flow and execution.	Tool commands, file reads, options, constraints application, reports. Ensures repeatable flow.	.tcl

8. Generated Output Files – Explanation

Output File	Description	Contents / Purpose	Format / Ext.
1. Gate-Level Netlist	Synthesized design netlist.	Gate-level connectivity, cells, instances, hierarchy. Input for Physical Design.	.v / .v.gz
2. Updated Constraints	Constraints after synthesis modifications.	Clock latencies, propagated delays, exceptions, generated clocks. Used in PD flow.	.sdc
3. Timing Reports	Timing analysis results.	WNS, TNS, setup / hold slack, path details, clock reports. Helps in timing closure.	.rpt / .rep
4. Area Reports	Area utilization details.	Total area, cell area, utilization, hierarchy breakdown.	.rpt
5. Power Reports	Power analysis results.	Total power, dynamic, leakage, power by hierarchy / block.	.rpt
6. Log / Messages	Tool run status and messages.	Warnings, errors, info, run summary. Useful for debugging.	.log

13. Viva / Interview Points

Q1. Why are PD input files important for synthesis ?
→ They provide design, technology, constraints and intent information required to generate optimized gate-level netlist that meets timing, power and area targets.

Q2. What happens if SDC constraints are missing or wrong ?
→ Tool may not optimize timing properly, leading to timing violations and failure to meet targets.

14. Summary Points

➢ PD input files include design, lib, SDC, UPF and TCL script.
➢ These inputs guide synthesis for timing, power and area.
➢ Missing or wrong inputs lead to wrong optimization and failures.
➢ Synthesis generates netlist, updated constraints and reports.
➢ Accurate inputs are crucial for smooth downstream PD flow.

PD Input Files

PD Input Files & Generated Output Files – Floor Planning

1. Introduction

Floorplanning is the first step of physical design. It defines the chip outline, core area, IO structure, macro placement, blockages and routing channels to achieve optimal PPA and congestion control.

2. Why It Is Needed ?

- Establishes physical architecture of the chip.
- Determines area utilization and routability.
- Guides placement, CTS and routing.
- Reserves resources for power, clock and IOs.
- Reduces late-stage ECOs and timing issues.

3. What If They Are Missed ?

- Poor or missing floorplan → high congestion.
- Inefficient area → large design, high cost.
- Timing closure becomes difficult.
- IR drop, EM and DRC issues increase.
- Routing failures and tape-out delays.

RISK

4. Advantages

- ✓ Optimized area utilization.
- ✓ Better routability & congestion control.
- ✓ Improved timing closure and power integrity.
- ✓ Smooth flow for placement, CTS and routing.
- ✓ Fewer ECOs and predictable implementation.

5. Disadvantages / Challenges

- x Incorrect macro placement → routing detours.
- x Wrong core size → timing / area issues.
- x Over or under utilization → PPA degradation.
- x Early estimation errors → rework / ECO.
- x Floorplan iterations consume time.

6. PD Input Files Summary

S.No.	Input File	Role	If Wrong / Missing
①	Gate Level Netlist	Describes logic, connectivity, hierarchy.	Synthesis mismatch, wrong area / timing.
②	Tech.lef (Technology LEF)	Tech info, layers, routing rules, vias, design rules.	Wrong rules, DRC violations.
③	Cells.lef (Std Cell / Macro LEF)	Cell / macro shapes, pins, obstruction, symmetry, site.	Wrong pin location, placement errors, DRC violations.
④	PNR SDC Constraints	Clocks, I/O delays, timing constraints, core sizing.	Timing violations, wrong channels / core sizing.

7. Explain Input Files

① Gate Level Netlist (.v / .sv)

- RTL synthesized logical netlist.
- Contains modules, instances, ports and connectivity.
- Used to estimate cell count, area and pin list.
- Hierarchy information is important for floorplan planning.

② Tech.lef (Technology LEF)

- Technology description file in LEF format.
- Contains layer informattion, via definitions, routing rules, design rules, track patterns, site definitions.
- Guides core area creation and routing resources.

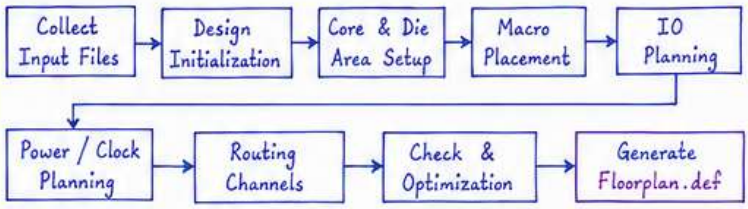
③ Cells.lef (Std Cell / Macro LEF)

- Contains physical abstracts of standard cells & macros.
- Includes size, pin locations, obstruction, blockages, symmetry, placement site, min spacing.
- Used for macro placement and obstruction creation.

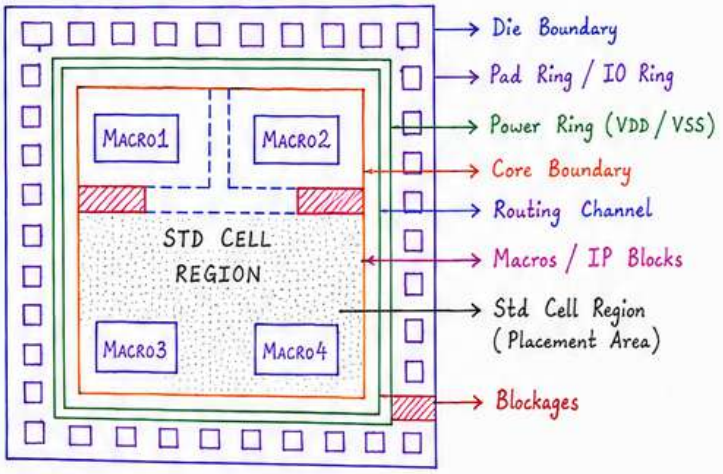
④ PNR SDC (Constraints)

- Physical design timing constraints.
- Clock definitions, I/O delays, false paths, multicycle paths, case analysis, timing exceptions.
- Used for core size estimation, channel width and analysis.

8. Overall Floorplanning Flow



9. Floorplan Components



10. Generated Output Files

Output File	Description	Content / Purpose	Format
Floorplan.def	Floorplan database after floorplanning.	• Die boundary. • Core boundary. • IO / Pad ring. • Macro placement. • Blockages. • Power & clock rings. • Routing channels.	.def

11. Comparison Table – Input vs Output

Aspect	Input Files	Generated Output
Purpose	Provide design, tech & constraints to tool.	Represents final floorplan setup.
Stage	Before floorplanning.	After floorplanning.
Examples	Netlist, Tech.lef, Cells.lef, PNR SDC.	Floorplan.def
Use	Guides tool decisions.	Input for power planning, placement & CTS.

12. Floorplanning Checks

- Core utilization (~60%-80%).
- Aspect ratio (~1:1 to 1:1.5).
- Macro keep-out & spacing.
- Power ring connectivity.
- Routing channel width.
- IO pin accessibility.
- DRC clean floorplan.

Key Point ☆

A good floorplan is the foundation for successful physical implementation. It decides the final PPA, routability and timing closure.

13. Viva / Interview Points

Q1. Why is floorplanning important in PD flow ?

→ It sets the physical architecture, controls congestion, area utilization and guides all downstream steps.

Q2. Which files are required for floorplanning ?

→ Gate level netlist, Tech.lef, Cells.lef and PNR SDC.

14. Summary Points

- ➢ Floorplanning defines the physical structure of the chip.
- ➢ Required inputs: Netlist, Tech.lef, Cells.lef and PNR SDC.
- ➢ Incorrect inputs lead to congestion, timing & DRC problems.
- ➢ Output floorplan.def is used for power planning, placement & CTS.
- ➢ Good floorplan ensures optimal PPA and smooth implementation.

PD Input Files

PD Input Files & Generated Output Files – Power Planning

1. Introduction

Power Planning creates a low resistance power delivery network (PDN) using power rings, straps and vias to distribute power (VDD) and ground (VSS) to the core. It ensures reliable operation by minimizing IR drop, EM violations and noise.

2. Why It Is Needed ?

- Provides robust power to all standard cells and macros.
- Reduces IR drop and ground bounce.
- Prevents electromigration (EM) failures.
- Ensures power integrity across the chip.
- Supports high current demand during switching.

3. What If It Is Missed ?

RISK

- High IR drop → timing failures, functional errors.
- EM violations → metal breakdown, reliability issues.
- Power rail noise → false switching / reset issues.
- Hotspots and overheating.
- Chip may fail in silicon.

4. Advantages

- ✓ Low IR drop and stable voltage.
- ✓ No EM and reliability issues.
- ✓ Better performance and timing closure.
- ✓ Supports high current and power density.
- ✓ Robust design for all operating corners.

5. Disadvantages / Challenges

- × Extra area overhead for rings, straps and vias.
- × More routing blockages.
- × Incorrect planning can cause congestion.
- × Over design → unnecessary area and cost.
- × Need accurate current estimation and analysis.

6. PD Input Files Summary

S.No.	Input File	Role	If Wrong / Missing
①	Floorplan.def	Defines die boundary, core area, macros, IOs and obstructions.	Incorrect region → PDN cannot be placed properly.
②	Gate Level Netlist	Required for power estimation and current calculation.	Wrong power numbers → IR drop/EM errors.
③	Tech.lef (Technology LEF)	Provides layer info, routing rules, min width, via info.	Wrong rules → DRC violations, wrong PDN design.
④	Cells.lef (Std Cell LEF)	Cell shapes, height, power pin locations and obstructions.	Pins not aligned → Power pins cannot be connected.
⑤	Liberty (.lib)	Cell power data, leakage, transition, capacitance info.	Wrong power values → bad current estimation.
⑥	UPF File	Power domains, supply nets, level shifters, isolation.	Domain info wrong → Power domain violations.
⑦	Current Estimation	Gives static, dynamic, peak & average current numbers.	Under-estimation → IR drop/EM failure. Over-estimation → over design.
⑧	Power Budget	Defines total power, IR drop target, EM limit, PPA goals.	No target → can't optimize PDN.

7. Explain Input Files

① Floorplan.def : Gives die size, core boundary, macro locations, IO ring and blockages. Power rings are placed along core boundary.

② Gate Level Netlist : Used to run power analysis to know current consumption of design (dynamic + leakage).

③ Tech.Lef : Contains layer stack, routing rules, min width/spacing, via definitions. Used to build valid PDN.

④ Cells.Lef : Gives power pin locations (VDD/VSS) on standard cells and macros. Alignment rows help connecting pins using straps.

⑤ Liberty (.lib) : Provides power numbers per cell (internal, switching, leakage, power states). Accurate current calculation depends on this.

⑥ UPF File : Defines supply nets, power domains, always-on/off domains, power switches, retention, isolation etc. Guides PDN connectivity.

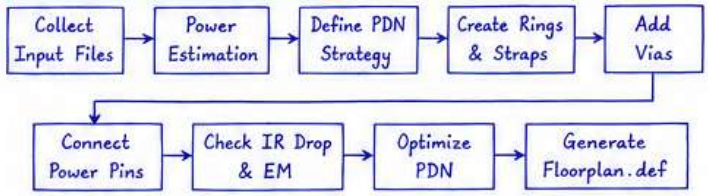
⑦ Current Estimation : From synthesis/PA tools. Includes static current, average dynamic current, peak current.

⑧ Power Budget : Target limits set by design team.
e.g., IR drop < 5%, EM < 80% limit, total power < X mW.

15. Summary Points

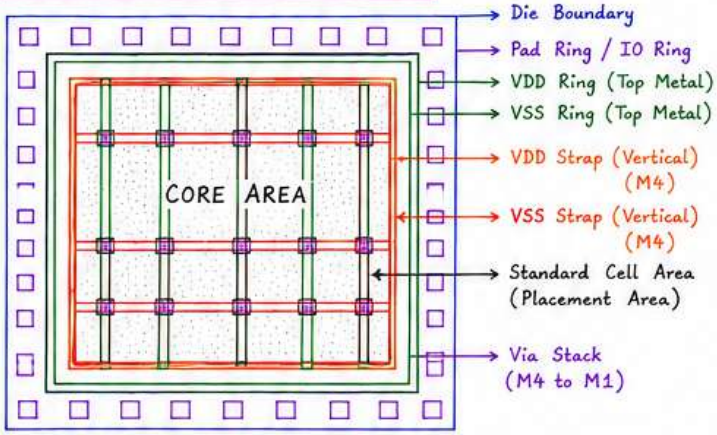
- ➤ Power Planning builds a robust PDN using rings, straps and vias.
- ➤ Accurate inputs (netlist, libs, tech, UPF, current) are critical.
- ➤ It reduces IR drop, noise and ensures reliability.
- ➤ Proper PDN improves timing, performance and silicon success.
- ➤ Generated Floorplan.def with PDN is used for placement.

8. Power Planning Flow

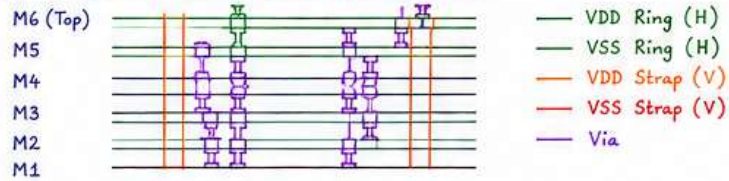


PDN Strategy decides ring width, strap width, spacing, via size and layers used.

9. Power Planning Concept Diagram



10. Typical PDN Structure (Multi Layer)



11. Generated Output File

Output File	Description	Content / Purpose	Format
Floorplan.def	Updated floorplan after power planning.	• Power rings. • Power straps. • Via locations. • Blockages update. • PDN connectivity.	.def

12. Comparison – Before vs After Power Planning

Aspect	Before Power Planning	After Power Planning
Power Delivery	Weak, high resistance path.	Strong, low resistance path.
IR Drop	High (may exceed limit).	Low (within target).
EM Risk	High chance of violation.	Within safe limit.
Noise	High ground bounce.	Reduced noise.
Timing	Unstable, errors possible.	Stable & reliable.
Reliability	Poor.	Good.

13. Power Planning Checks

- IR drop check (Static / Dynamic).
- EM check (Current density).
- Via density check.
- Power pin connectivity check.
- IR/EM across all corners.
- Redundant via rule.
- Metal density check.
- Domain connectivity (UPF).

Key Point ☆

A good power plan ensures low IR drop, no EM violations and stable power delivery across all corners.
It is the backbone of a reliable chip.

14. Viva / Interview Points

Q1. What is IR drop ?
→ Voltage drop in power network due to finite resistance.

Q2. What is EM ?
→ Electromigration is metal degradation due to high current density.

Q3. Why power rings are placed on top metal layers ?
→ Top metals have lower resistance and better current capacity.

Q4. What is the goal of power planning ?
→ Low IR drop, no EM issues and reliable power delivery.

PD Input Files

PD Input Files & Generated Output Files – Placement

1. Introduction

Placement determines the physical locations of all standard cells and macros within the core area defined by floorplan. Goal is to achieve good timing, low congestion, less wirelength, balanced utilization and easy routing.

2. Why It Is Needed ?

- Transforms abstract netlist into physical locations.
- Minimizes wirelength and timing violations.
- Reduces congestion and routing detours.
- Improves performance, power and area.
- Prepares design for CTS and routing.

3. What If It Is Missed ?

RISK



- High congestion → routing failures.
- Long interconnect → timing failures.
- Poor timing closure and high power.
- More ECOs and tape-out delay.

4. Advantages

✓ Better timing and shorter critical paths.
✓ Lower wirelength and improved routability.
✓ Balanced area utilization.
✓ Less congestion and predictable routing.
✓ Good starting point for CTS and routing.

5. Disadvantages / Challenges

× Congestion in some regions may remain.
× Timing optimization may need iterations.
× Large designs → high runtime and memory.
× Poor constraints → wrong placement results.

6. PD Input Files Summary

S.No.	Input File	Role	If Wrong / Missing
①	Floorplan.def	Defines die, core area, IOs, macros, blockages, rows and sites.	Wrong or incomplete def → placement errors or overlap.
②	Synthesised Netlist	Logical connectivity of the design.	Missing / wrong netlist → wrong connectivity.
③	Tech. lef	Technology layers, routing rules, via rules, design rules.	Wrong tech → DRC violations, invalid placement.
④	Cells. lef	Physical abstracts of std cells / macros, pin locations, obstructions.	Wrong cell info → pin mismatch, overlap, wrong cell orientation.
⑤	SDC Constraints	Clocks, I/O delays, timing constraints, exceptions.	Wrong constraints → poor optimization and timing failure.

7. Explain Input Files

① **Floorplan.def**
- Contains die size, core boundary, rows, sites, macros, IOs, blockages and routes (if any).
- It is the starting physical boundary for placement.

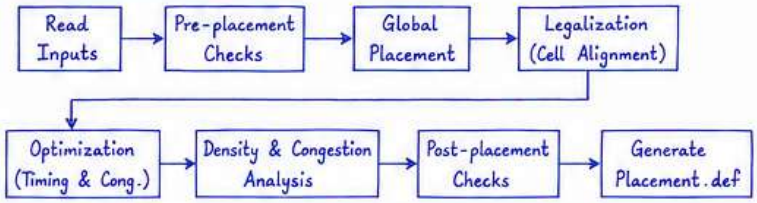
② **Synthesised Netlist (Gate Level)**
- RTL synthesised gate-level netlist.
- Contains instances, modules, ports and connectivity.
- Used to build connectivity graph for placement.

③ **Tech.LEF (Technology LEF)**
- Layer information, routing rules, via definitions, design rules.
- Manufacturing grids, track patterns and constraints.

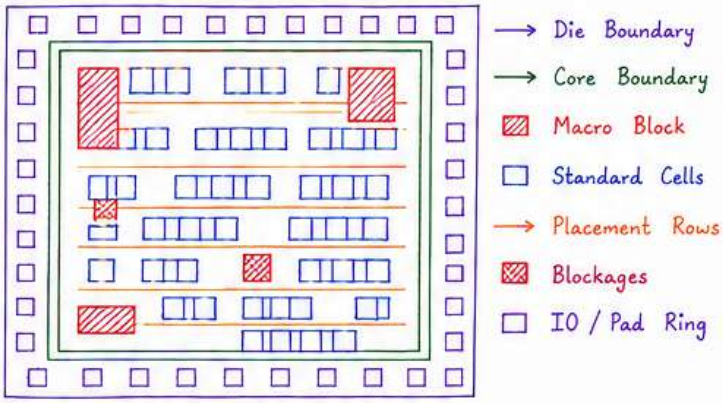
④ **Cells.LEF (Std Cell / Macro LEF)**
- Physical models of standard cells and macros.
- Pin locations, obstructions, blockages, symmetry,
- site type and cell size.

⑤ **SDC Constraints**
- Specifies clocks, uncertainties, I/O delays, false paths, multicycle paths, case analysis, timing groups.
- Guides timing-driven placement.

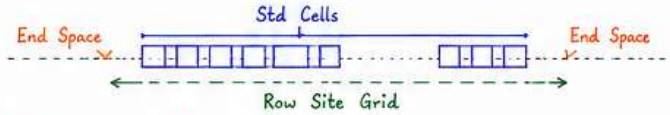
8. Placement Flow



9. Placement Concept Diagram



10. Typical Placement Row Structure



11. Generated Output File

Output File	Description	Content / Purpose	Format
Placement.def	Placement database after placement.	• Placed std cells & macros. • Cell coordinations. • Orientations. • Updated net connectivity. • Row information.	.def

12. Comprarison – Before vs After Placement

Aspect	Before Placement	After Placement
Design State	Netlist + Floorplan	Cells are placed in rows
Wirelength	High (unknown)	Reduced wirelength
Timing	Unknown	Better timing, less violations
Congestion	Not known	Congestion reduced
Utilization	Unbalanced	Balanced utilization
Next Step	Placement not done	Ready for CTS

13. Placement Checks

- Utilization (Target ~ 60% – 80%).
- Density distribution.
- Congestion (Global / Detailed).
- Timing (setup / hold).
- DRC & site alignment.

Key Point ☆

Good placement reduces wirelength, improves timing, decreases congestion and makes routing easier.

14. Viva / Interview Points

Q1. Why is placement important in Physical Design ?
→ It determines cell locations to optimize timing, congestion, wirelength and area before routing.

Q2. What happens if placement is poor ?
→ High congestion, long wires, timing failures, high power and routing may fail.

Q3. Which inputs are required for placement ?
→ Floorplan.def, Netlist, Tech.lef, Cells.lef and SDC.

15. Summary Points

➤ Placement places all std cells and macros in rows of core.
➤ Required inputs: Floorplan.def, Netlist, Tech.lef, Cells.lef, SDC.
➤ Good placement reduces wirelength, congestion and timing violations.
➤ Placement.def is generated and used for CTS and routing.
➤ Proper constraints and LEF accuracy are crucial for success.

PD Input Files

PD Input Files & Generated Output Files – Clock Tree Synthesis

1. Introduction

Clock Tree Synthesis (CTS) builds a clock distribution network that delivers the clock from source to all sequential elements with controlled skew, latency and uncertainty.
CTS ensures timing closure and reliable operation.

2. Why It Is Needed ?

- Distributes clock to all sequential elements.
- Reduces clock skew and insertion delay.
- Balances clock arrival times.
- Improves timing, performance and power.
- Prevents hold violations due to skew.

3. What If It Is Missed ?

RISK

- Large clock skew → setup/hold violations.
- Unbalanced latency → timing failures.
- High dynamic power due to glitch & buffers.
- Less reliability and functional failures.
- Chip may not meet timing sign-off.

4. Advantages

✓ Balanced clock arrival → better timing.
✓ Lower skew and uncertainty.
✓ Reduced power by optimized buffer insertion.
✓ Predictable and reliable clock distribution.
✓ Supports high frequency operation.

5. Disadvantages / Challenges

× CTS can increase area (buffers, wires).
× Too much buffering → higher power.
× Incorrect constraints → wrong skew targets.
× Iterative process with placement & routing.
× More complex in multi-clock and multi-voltage domains.

6. PD Input Files Summary

S.No.	Input File	Role	If Wrong / Missing
①	Placement.def	Def of design after placement. Contains cell locations, rows, blockages, nets.	Wrong or incomplete DEF → wrong clock topology / overlaps.
②	Synthesized Netlist	Logical connectivity (sequential cells, hierarchy).	Missing / wrong netlist → wrong connectivity.
③	Tech.lef	Tech info, layers, routing rules, via defs.	Wrong tech → DRC violations, invalid CTS design.
④	Cells.lef	Std cell abstracts for buffers, inverters, drives, pins.	Wrong cell info → bad buffer insertion and timing.
⑤	SDC Constraints	Clocks, I/O delays, latencies, uncertainty, exceptions.	Wrong constraints → wrong skew target & timing violations.

7. Explain Input Files

① **Placement.def**
- Contains the physical database after placement.
- Includes rows, sites, cell coordinates, blockages, obstructions and netlist connectivity.
- Used as the starting point for CTS.

② **Synthesized Netlist**
- Gate level netlist from synthesis.
- Contains all sequential elements (FFs, latches) and logic. CTS identifies clock sinks from this netlist.

③ **Tech.lef (Technology LEF)**
- Defines routing layers, via rules, track patterns, Manufacturing grids, min spacing, etc.
- Guides buffer placement and routing of clock tree.

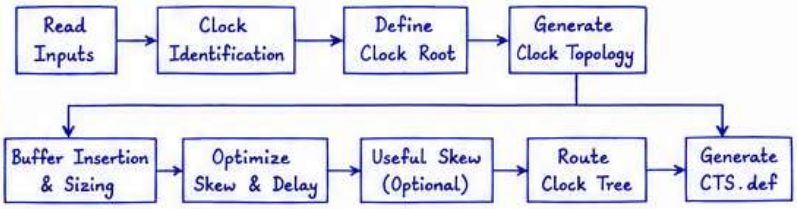
④ **Cells.lef (Std Cell LEF)**
- Physical abstracts of standard cells.
- Provides pin locations, driving strengths of buffers, inverters and clock gates.

⑤ **SDC Constraints**
- Defines clock definitions, latencies, uncertainties, false paths and exceptions.
- Specifies max skew targets and useful skew if any.

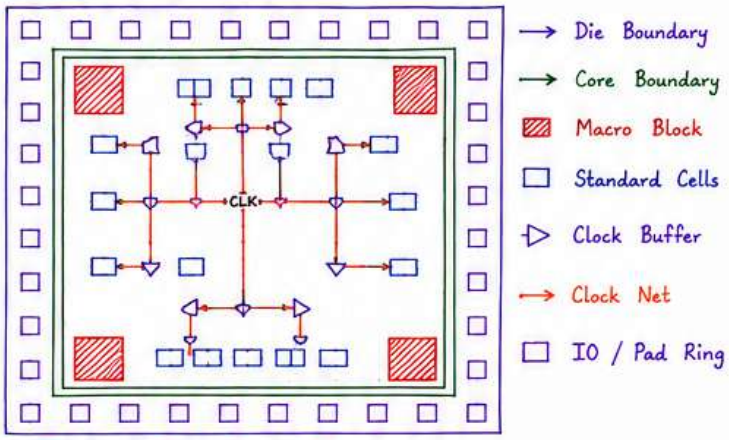
15. Summary Points

➢ CTS builds the clock distribution network after placement.
➢ Required inputs: Placement.def, Netlist, Tech.lef, Cells.lef and SDC.
➢ CTS reduces clock skew and improves timing reliability.
➢ CTS.def is generated with clock buffers and clock routing.
➢ Proper constraints and LEF accuracy are essential for success.

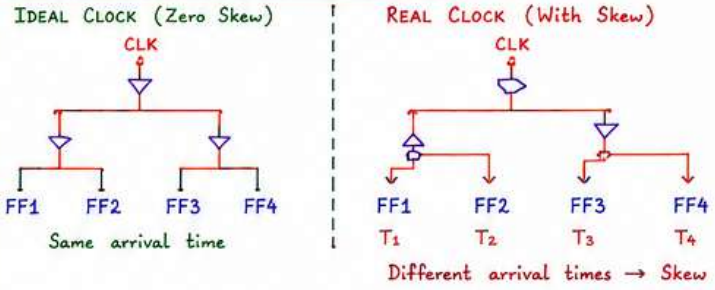
8. CTS Flow



9. CTS Concept Diagram



10. Ideal vs Real Clock Distribution



11. Generated Output File

Output File	Description	Content / Purpose	Format
CTS.def	DEF database after Clock Tree Synthesis.	• Inserted clock buffers. • Clock tree wires. • Updated net connectivity. • Clock root and sinks. • Blockages (if any).	.def

12. Before vs After CTS

Aspect	Before CTS (Placement)	After CTS
Clock Source	Only clock port / root	Clock root + clock tree
Clock Net	Not distributed	Distributed to all sinks
Clock Skew	High / Uncontrolled	Controlled within target
Insertion Delay	N/A	Balanced across paths
Timing	Many violations	Reduced violations
Power	Lower (no buffers)	Optimized (acceptable)
Design State	Placement.def	CTS.def

13. CTS Checks

- Clock skew (Target < skew limit).
- Insertion delay and latency.
- Max transition on clock nets.
- Capacitance and buffer loading.
- DRC clean clock tree.
- Clock reconvergence pessimism (CRP).

Key Point ☆
CTS creates a balanced clock distribution network with controlled skew and latency to achieve timing closure and reliability.

14. Viva / Interview Points

Q1. Why is CTS important in Physical Design ?
→ It balances clock arrival times and controls skew, latency and uncertainty to achieve timing closure.
Q2. What happens if clock skew is high ?
→ Setup / hold violations occur leading to functional failures.
Q3. Which files are required for CTS ?
→ Placement.def, Netlist, Tech.lef, Cells.lef and SDC Constraints.

PD Input Files

PD Input Files & Generated Output Files – Routing

1. Introduction

Routing connects all the signals physically using metal layers after CTS. It converts logical connectivity into real interconnects while meeting DRC, timing, IR drop and reliability requirements.

2. Why It Is Needed ?

- Creates the final physical connections between cells.
- Ensures electrical connectivity of the design.
- Reduces parasitics and improves timing.
- Meets design rules and manufacturability.
- Avoids opens, shorts and floating nets.

3. What If It Is Missed ?

RISK

- Design is not electrically connected.
- Open nets → functional failure.
- High resistance / long wires → timing failure.
- Violates DRC → tape-out not possible.
- High IR drop, crosstalk and reliability issues.

4. Advantages

✓ Complete signal connectivity.
✓ Better timing and reliable operation.
✓ Optimized wirelength and congestion control.
✓ Meets DRC and antenna rules.
✓ Improved performance and power.

5. Disadvantages / Challenges

✗ High congestion → routing detours.
✗ More metal layers → higher capacitance.
✗ Crosstalk and noise issues.
✗ IR drop and EM (electromigration) concerns.
✗ Iterative ECOs increase runtime and cost.

6. PD Input Files Summary

S.No.	Input File	Role	If Wrong / Missing
①	CTS.def	DEF database after CTS (cells, clock tree, blockages, vias, pins).	Missing nets, wrong blockages → routing errors, opens/shorts.
②	Routing Constraints .tcl	Tool commands & rules for routing (layers, widths, spacing, via rules, cost, priority).	Wrong rules → DRC violations, congestion, wrong routing.

7. Explain Input Files

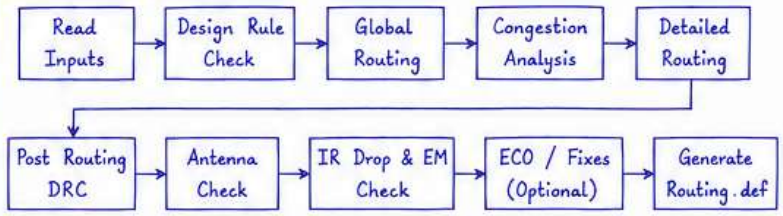
① CTS.def
- DEF file generated after Clock Tree Synthesis.
- Contains placed cells, macros, clock tree, IOs, blockages, rings, pins, vias, nets and design boundaries.
- Serves as the starting physical database for routing.

② Routing.constraints .tcl
- Tcl script with routing rules and tool commands.
- Defines: layer stack, preferred directions, min width, min spacing, via rules, via stacking, target utilization, cost factors, antenna rules, shielding rules etc.
- Guides the router to produce DRC clean and optimized design.

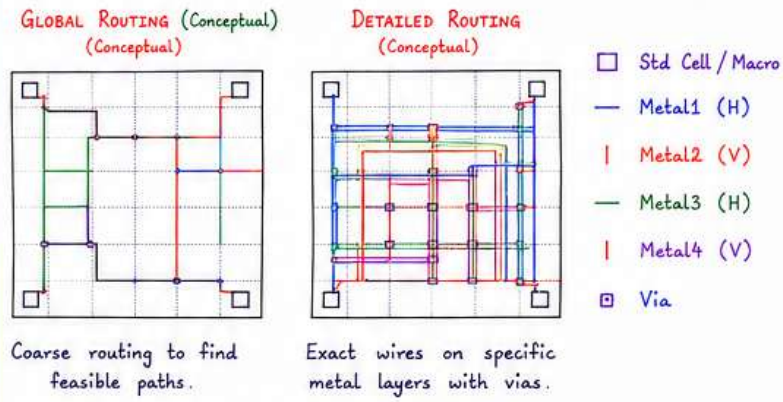
15. Summary Points

➢ Routing establishes the final physical interconnects.
➢ Required inputs are CTS.def and Routing Constraints .tcl.
➢ Global routing finds path & Detailed routing builds wires.
➢ Routing.def is generated after successful routing.
➢ DRC clean routing ensures manufacturable chip.
➢ Optimize for timing, congestion, IR drop and reliability.

8. Routing Flow



9. Routing Concept Diagram



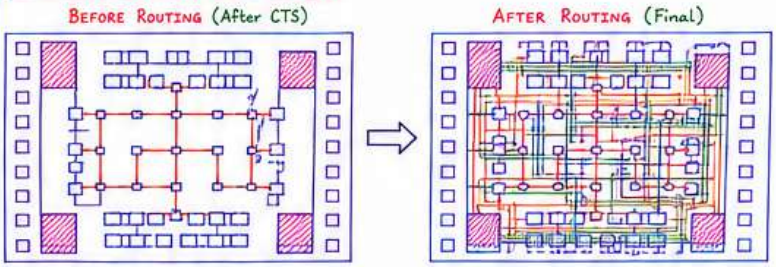
10. Global vs Detailed Routing

Aspect	Global Routing	Detailed Routing
Purpose	Finds feasible paths on coarse grid.	Builds exact wires on actual layers.
Resolution	Coarse (GCells).	Fine (Tracks).
Output	Guidance / Routes.	Actual routed wires.
DRC	DRC not guaranteed.	DRC clean.
Congestion	Identifies & reduces.	Resolves remaining.
Runtime	Faster.	Relatively slower.

11. Generated Output File

Output File	Description	Content / Purpose	Format
Routing.def	Final DEF database after routing.	• Routed nets and wires. • Via locations. • Updated connectivity. • Routed blockages (if any).	.def

12. Before vs After Routing



13. Routing Checks

- DRC (Design Rule Check).
- Antenna rule check.
- IR Drop analysis.
- Electromigration (EM) analysis.
- Timing analysis (setup / hold).
- Net connectivity and opens check.

Key Point ☆

Quality routing ensures DRC clean connectivity, optimized timing, low IR drop and high reliability for successful tape-out.

14. Viva / Interview Points

Q1. Why is routing the last major step in PD flow ?
→ Because it creates actual physical interconnects after CTS.

Q2. What is the difference between global and detailed routing ?
→ Global routing finds path on coarse grid, detailed routing builds exact wires on metal layers.

Q3. Which files are required for routing ?
→ CTS.def and Routing Constraints .tcl.

PD Input Files

— Sign Off —

1. Introduction

Sign-off stage is the final physical verification step in PD flow. It ensures that the design meets all sign-off criteria in terms of timing, physical correctness, power integrity and manufacturability before tape-out.

2. Why Sign Off Is Needed ?

- To verify design is manufacturable and reliable.
- To ensure timing closure with extracted parasitics.
- To check physical rule compliance (DRC, LVS, ERC etc).
- To avoid costly respins and silicon failures.

3. What If Sign Off Is Missed ?

- ✗ Design may have timing violations → functional failure.
- ✗ DRC/LVS errors → foundry rejects the layout.
- ✗ Power issues (IR drop, EM) → reliability problems.
- ✗ Antenna violations → oxide damage during fabrication.
- ✗ Extra cost, more time and project delay.

4. Advantages

- ✓ Ensures correct and manufacturable physical layout.
- ✓ Improves design reliability and silicon success.
- ✓ Reduces post-silicon failures and risk.
- ✓ Better PPA (Power, Performance, Area) closure.

5. Disadvantages / Limitations

- Sign-off takes high runtime and CPU resources.
- Requires correct setup and clean input data.
- Large designs need huge storage and memory.
- Any missed violation → costly iteration.

6. PD Input Files Used For Sign-Off

S.No	Input File	Description / Role
1.	Routing.def	Final placed and routed design in DEF format. • Contains final geometry, nets, vias. • Used for all sign-off checks.
2.	SPEF (RC Extracted Parasitics)	Standard Parasitic Exchange Format. • Contains RC parasitics (R, C, Coupling). • Used by STA for accurate timing analysis.
3.	.Lib (Timing Library)	Standard cell timing library. • Contains cell delays, constraints, pin info, slew, load tables. • Used by STA for timing calculation.
4.	.sp (SPICE Netlist)	Extracted SPICE netlist for circuit level verification. • Used for analog / mixed-signal checks. • Helps in detailed circuit validation.
5.	DRC (.tcl)	DRC rule deck in TCL format. • Contains foundry design rules. • Used by DRC tool to check geometry.
6.	Rcx (.tcl) (LVS/Connectivity Rules)	Rule deck for LVS connectivity check. • Contains device recognition rules. • Used by LVS tool to compare layout vs schematic.

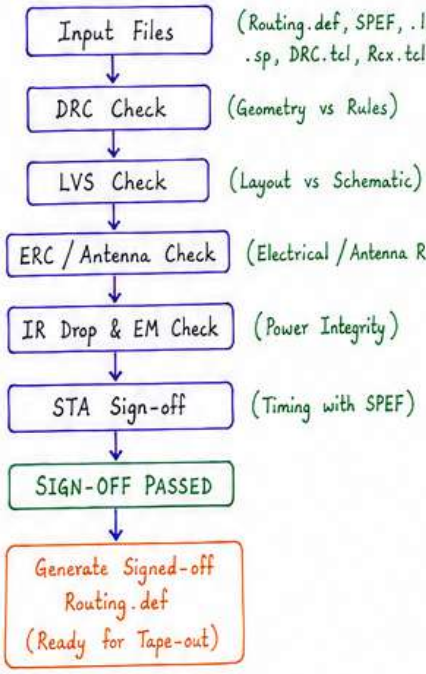
7. Sign-Off Checks Performed

✓ Timing Sign-off (Setup / Hold)	✓ Electromigration (EM) Check
✓ DRC (Design Rule Check)	✓ Antenna Check
✓ LVS (Layout Vs Schematic)	✓ Power Intent Check
✓ ERC (Electrical Rule Check)	✓ Physical Verification
✓ IR Drop Analysis	✓ STA with Extracted RC

Key Point

Sign-off ensures the design is CORRECT + MANUFACTURABLE + RELIABLE before it is sent for tape-out. No violations should remain at sign-off.

8. Sign-Off Flow (Conceptual)



Output Generated

- Routing.def (Signed-off) → Final verified layout database.
- This DEF is used for GDSII generation and final tape-out.

9. Input Files vs Their Usage

File	Type	Main Usage in Sign-Off	Tool Used
Routing.def	Layout Database	Contains final routed design for all checks.	Calibre, StarRC, Tempus, etc.
SPEF	Parasitics	Used in STA for accurate timing sign-off.	StarRC, Tempus
.Lib	Timing Library	Provides cell timing, constraints for STA.	Tempus, PrimeTime
.sp	Circuit Netlist	Used for circuit verification, LVS & custom checks.	HSPICE, Spectre, Calibre xRC
DRC (.tcl)	Rule Deck	Checks design rules, spacing, width, enclosure, etc.	Calibre, Pegasus
Rcx (.tcl)	LVS Rule Deck	Checks connectivity between layout and schematic.	Calibre LVS

10. Importance of Each File

- → Routing.def → Backbone file for physical verification.
- → SPEF → Ensures accurate timing with real parasitics.
- → .Lib → Necessary for correct timing evaluation.
- → .sp → Used for deeper circuit level validation.
- → DRC (.tcl) → Ensures design follows foundry rules.
- → Rcx (.tcl) → Ensures layout matches the schematic.

Without Proper Sign-off

- → Chip may fail in silicon validation.
- → Foundry may reject the layout.

Practical Observation

- Most violations are found in early sign-off iterations.
- Each iteration must clean all critical violations.
- Final sign-off must be fully clean.
- Signed-off Routing.def is the golden database for tape-out.

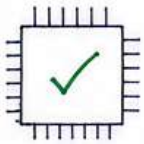
Viva / Interview Q&A

Q. Why sign-off is the most critical stage ?
→ Because any missed violation can cause silicon failure and huge re-spin cost.

Q. Which file is used for timing sign-off ?
→ SPEF (Parasitic) + .Lib (Timing Library) are used by STA tool.

11. Summary (Revision Points)

1. Sign-off is the final verification step before tape-out.
2. Input files: Routing.def, SPEF, .Lib, .sp, DRC(.tcl), Rcx(.tcl).
3. Checks performed: Timing, DRC, LVS, ERC, IR Drop, EM, Antenna, etc.
4. All checks must be 100% clean (no violations).
5. Output after sign-off → Signed-off Routing.def (verified layout).
6. This verified DEF is used for GDSII generation and tape-out.
7. Sign-off ensures manufacturable, reliable and correct silicon.



SIGN-OFF CLEAN !!

PD Input Files

GDSII Export

1. Introduction

PD tools require various input files to perform floorplanning, placement, CTS, routing, extraction and sign-off. These files describe design, technology, libraries and physical intent required to build a manufacturable layout.

2. Why Input Files are Needed ?

- Define design connectivity, timing and physical intent.
- Provide technology rules, cell information, constraints.
- Enable accurate placement, routing and verification.
- Achieve timing closure, low power and DRC clean sign-off.

3. What if Input Files are Missed ?

- ✗ Wrong or incomplete info → placement / routing failures.
- ✗ Timing violations due to missing constraints / libraries.
- ✗ DRC / LVS failures and tape-out got delayed.
- ✗ Increased iterations → more cost and schedule slip.

4. Advantages

- ✓ Accurate and consistent physical implementation.
- ✓ Improves design quality and sign-off confidence.
- ✓ Reduces iterations and overall development time.
- ✓ Ensures manufacturable and reliable silicon.

5. Disadvantages / Limitations

- Large number of files → management complexity.
- Wrong versions → major functional or timing issues.
- Tool dependency and format compatibility issues.
- Requires good understanding of file roles.

6. PD Input Files - Details

Input File	Description / Role
1. Netlist (Verilog/VHDL)	Synthesized gate-level netlist. Defines connectivity between cells.
2. SDC (Constraints)	Timing constraints, clocks, I/O delays, false paths, multicycle paths etc.
3. Library Files	• LIB (Timing lib) → cell timing, power, area, pin info. • NLDM/CCS → for accurate timing.
4. LEF Files	Technology LEF (tech.lef) → layers, vias, design rules. Cell LEF (stdcell.lef, macro.lef) → cell shapes, pins, obstractions.
5. DEF (Logical DEF)	Design floorplan, die boundary, macro locations, regions, routing blockages, component placement (pre-placement info).
6. MAP / Tech Files	Layer mapping, routing tracks, via rules, design rules.
7. Power Intent (UPF/CPF)	Power domains, power switch, power states, isolation, retention.
8. IO Files	IO pad locations, IO constraints, pin orientations.
9. SPEF (Optional)	Back-annotated parasitics (for physical timing analysis).
10. Other Files	Scripts (TCL), tool settings, DRC rule decks, Antenna rules.

7. Key Input Files Explained

A) Verified Routed .DEF File

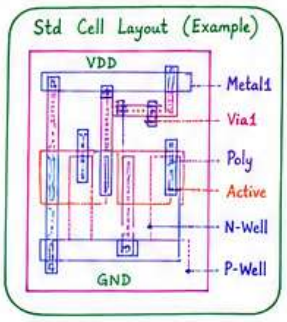
- Generated after detailed routing and physical verification.
- Contains complete placed and routed design.
- Used for extraction and sign-off.
- Key information in Verified Routed.def :
  - → Design name, units, die area.
  - → Component (cells, macros) locations.
  - → Pins, nets, routing (wires and vias).
  - → Special nets (VDD, VSS, clocks).
  - → Track grids and blockages.
  - → Routing layers used.
  - → Regions, groups and constraints.

.DEF (Snippet)

```
VERSION 5.8 ;
DESIGN top-design ;
UNITS DISTANCE MICRONS 2000;
DIEAREA (0 0) (2000 2000) ;
COMPONENTS 2 ;
- U1 NAND2_X1 +
  PLACED (100 150) FS ;
- U2 INV_X1 +
  PLACED (300 150) FS ;
END COMPONENTS
PINS 2 ;
- A + NET A + DIRECTION INPUT;
- Y + NET Y + DIRECTION OUTPUT;
END PINS
NETS 10 ;
... (nets and routing info)
END NETS
END DESIGN
```

B) Standard Cell GDS / OASIS File

- Contains the physical layout shapes of standard cells.
- Provided by Foundry / Library vendor.
- Used by tools for placement, DRC, extraction.
- Two common formats :
  - → GDSII → widely used layout format.
  - → OASIS → compressed, supports larger designs.
- Includes :
  - → Layout shapes on multiple layers.
  - → Pin shapes and labels.
  - → Cell boundary.
  - → Text labels, obstructions.



8. Output File Generated

Verified Routed.def + Std Cell GDS / OASIS → **GDSII** (Final Layout)

9. GDSII Export

- Final step before tape-out.
- Converts the internal layout database into GDSII format.
- Contains all layers (routing, vias, implants, labels etc.).
- Submitted to Foundry for fabrication.

10. Why GDSII is Needed ?

- Industry standard for IC fabrication.
- Communicates exact layout to Foundry.
- Used for Mask generation.

Key Point

★ Proper and correct input files ensure smooth Physical Design flow and successful tape-out.

11. What if GDSII is Missed ?

- ✗ Foundry cannot fabricate the chip.
- ✗ Tape-out cannot be completed.
- ✗ Project delay and cost increase.

12. Advantages of GDSII Export

- ✓ Industry standard.
- ✓ Accurate layout delivery.
- ✓ Ensures correct mask generation.
- ✓ Enables smooth tape-out.
- ✓ Supports multi-project and ECO flow.

13. Summary (Revision Points)

1. PD needs multiple input files for correct implementation.
2. Missing or wrong inputs cause timing, DRC and routing issues.
3. Verified Routed.def contains placed and routed information.
4. Standard Cell GDS/OASIS provides cell physical layouts.
5. These inputs produce GDSII file for tape-out.
6. GDSII is required for mask generation and fabrication.
7. Correct inputs → better quality, less iterations and early tape-out.

Practical Observation

- Always use latest and consistent versions of input files.
- Verify input files before starting implementation.
- Good input quality = good output quality.

Interview / Viva

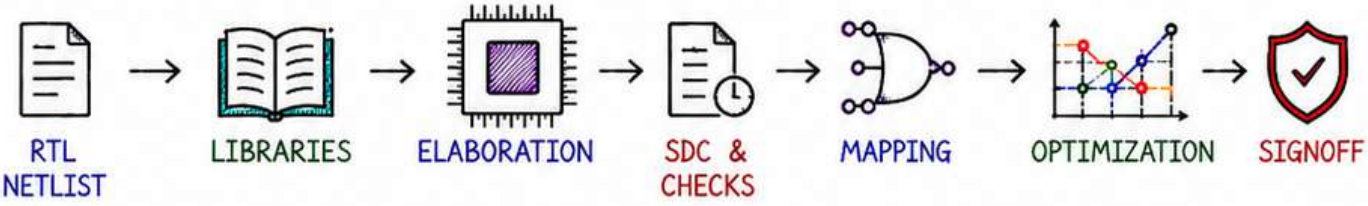
Q. What is the role of DEF file ?
→ It describes design floorplan, components, nets and routing information.

Q. Why is GDSII important ?
→ It is required for fabrication and mask generation in foundry.

Sign-off Checklist (Outputs)

✓ DRC Clean ✓ LVS Clean ✓ Antenna Clean
✓ Timing Clean ✓ IR Drop OK ✓ EM OK
✓ Power OK ✓ Physical Verification Clean

All checks pass → Ready for Tape-out !



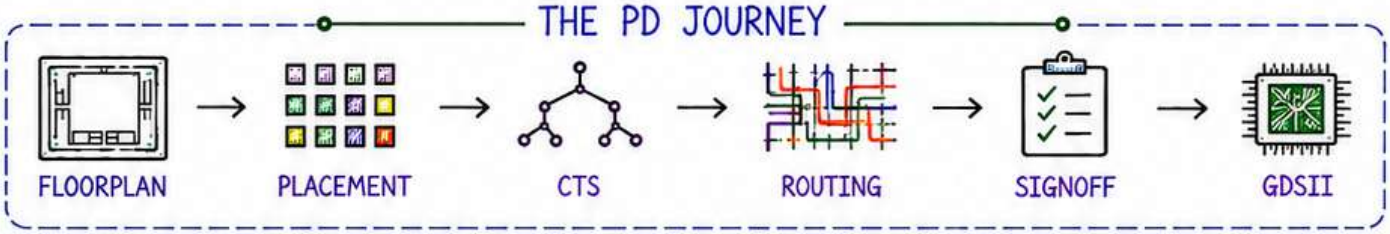


THANK You!

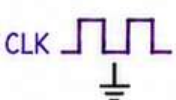
Your time, effort and attention
made this learning journey
meaningful and successful.

KEEP EXPLORING, KEEP INNOVATING,
KEEP BUILDING THE FUTURE !

> "From Design Intent to Silicon Success –
> Every Step in Physical Design Matters."



Thank you for being a part of this learning path.
Hope these notes help you in mastering
Physical Design and achieving your goals.



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